

Rise 120° of cam rotation with simple harmonic motion (SHM); then
 Dwell for 60° of cam rotation; then
 Fall 90° of cam rotation in simple harmonic motion.
 Fall : 90° of cam rotation with simple harmonic motion (SHM); then
 Dwell for remaining period. (CO1)

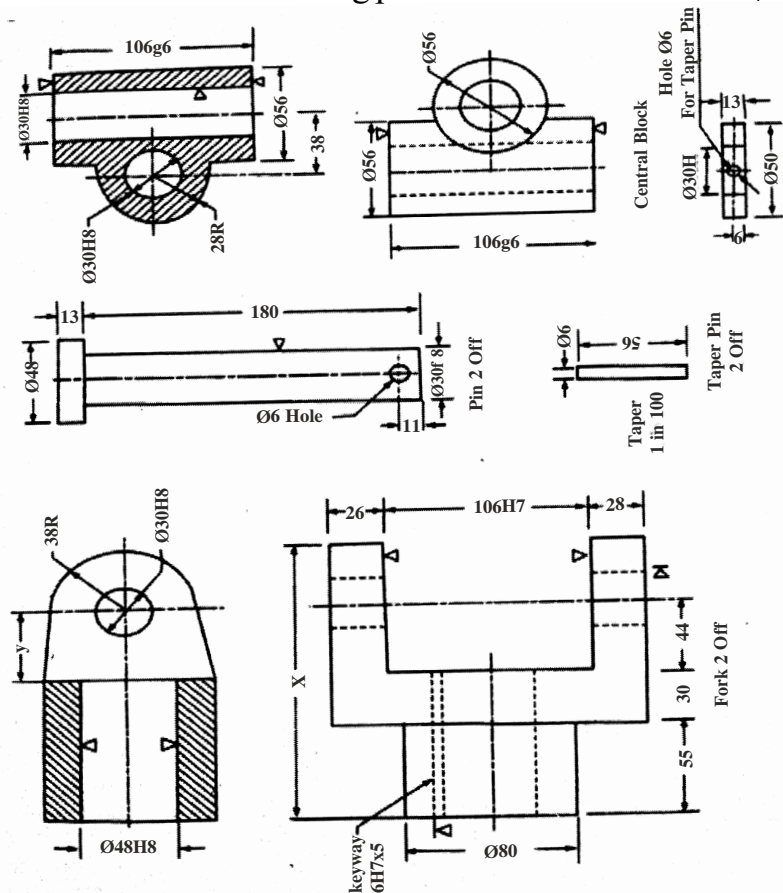


Fig. 1: Detail drawing of Universal Joint
 Dimension Units may be taken in mm.

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4th Sem / Auto

Subject:- Automobile Engineering Drawing

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 When the tolerance distribution is on both sides of the basic size, it is known as (CO3)
 a) Unilateral tolerance b) Compound tolerance
 c) Bilateral tolerance d) Geometric tolerance
- Q.2 If the largest permissible diameter of the shaft is greater than the diameter of the smallest hole, then the fit is called as (CO3)
 a) Clearance Fit b) Transition fit
 c) Interference fit d) None of the above
- Q.3 The algebraic difference between the minimum limit of size and its corresponding basic size is known as (CO3)
 a) upper deviation b) Upper limit
 c) Lower deviation d) Lower limit
- Q.4 The difference between the maximum and minimum limits of size is known as (CO3)
 a) Allowance b) Limits of size
 c) Tolerance d) Clearance
- Q.5 In Hole basis system (CO3)
 a) The size of shaft is kept constant and the size of hole is variable

(180)

(4)

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(1)

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- b) The size of hole is kept constant and the size of shaft is variable
- c) The sizes of both the hole and the shaft are constant
- d) The sizes of both the hole and the shaft are variable

Q.6 Bearing is used to (CO3)

- a) Support a shaft b) Transfer power
- c) Connect two shafts d) All of the above

Q.7 A spark plug has (CO3)

- a) One electrode b) Three electrodes
- c) Two electrodes d) No Electrode

Q.8 Which of the following is not a part of leaf spring:

- a) Eye b) Main leaf (CO3)
- c) U-bolt d) Small end

Q.9 Which of the following are not the components of overhead valve mechanism: (CO3)

- a) Cam & Cam shaft
- b) Valve guide & valve stem
- c) Cam follower & rocker arm
- d) Roller & cage

Q.10 A wheel cylinder consists of: (CO3)

- a) One cup b) One piston
- c) Two push rods d) Two cylinders

SECTION-B

Note: Very short answer type questions; attempt any five:
(5x2=10 marks)

Q.11 Define the base circle of a cam. (CO2)

Q.12 Write any two types of cam motions. (CO2)

Q.13 Define dwell in case of cam. (CO2)

Q.14 Write the relation between diametral pitch and pitch circle diameter. (PCD). (CO2)

Q.15 Draw a sketch of slip joint. (CO1)

Q.16 Name the two types of shoes of drum brake. (CO2)

SECTION-C

Note: Draw freehand sketches of any two: (2x10=20)

Q.17 Camshaft (CO1)

Q.18 Wheel cylinder (CO1)

Q.19 Battery ignition system (CO1)

Q.20 Overhead valve mechanism (CO1)

SECTION-D

Note: Attempt any two questions: (2x30=60)

Q.21 Draw following views of four stroke petrol engine by taking appropriate dimensions:

- a) Front view, full in section
- b) Top view
- c) side, view, left half in section (CO1,2)

Q.22 Fig. 1 shows the detailed drawing of a universal Joint. Draw the following orthographic views:

- a) Full sectional front view
- b) Top view (CO1,2)

Q.23 Draw the profile of involute teeth by Prof. Unwin's method having 24 teeth, module pitch 8 mm and pressure angle 22°. Draw at least three teeth and label all the circles. (CO1)

Q.24 Draw the cam profile for a radial knife edge follower having following conditions:

- Lift = 50 mm;
- Base circle radius = 30 mm;